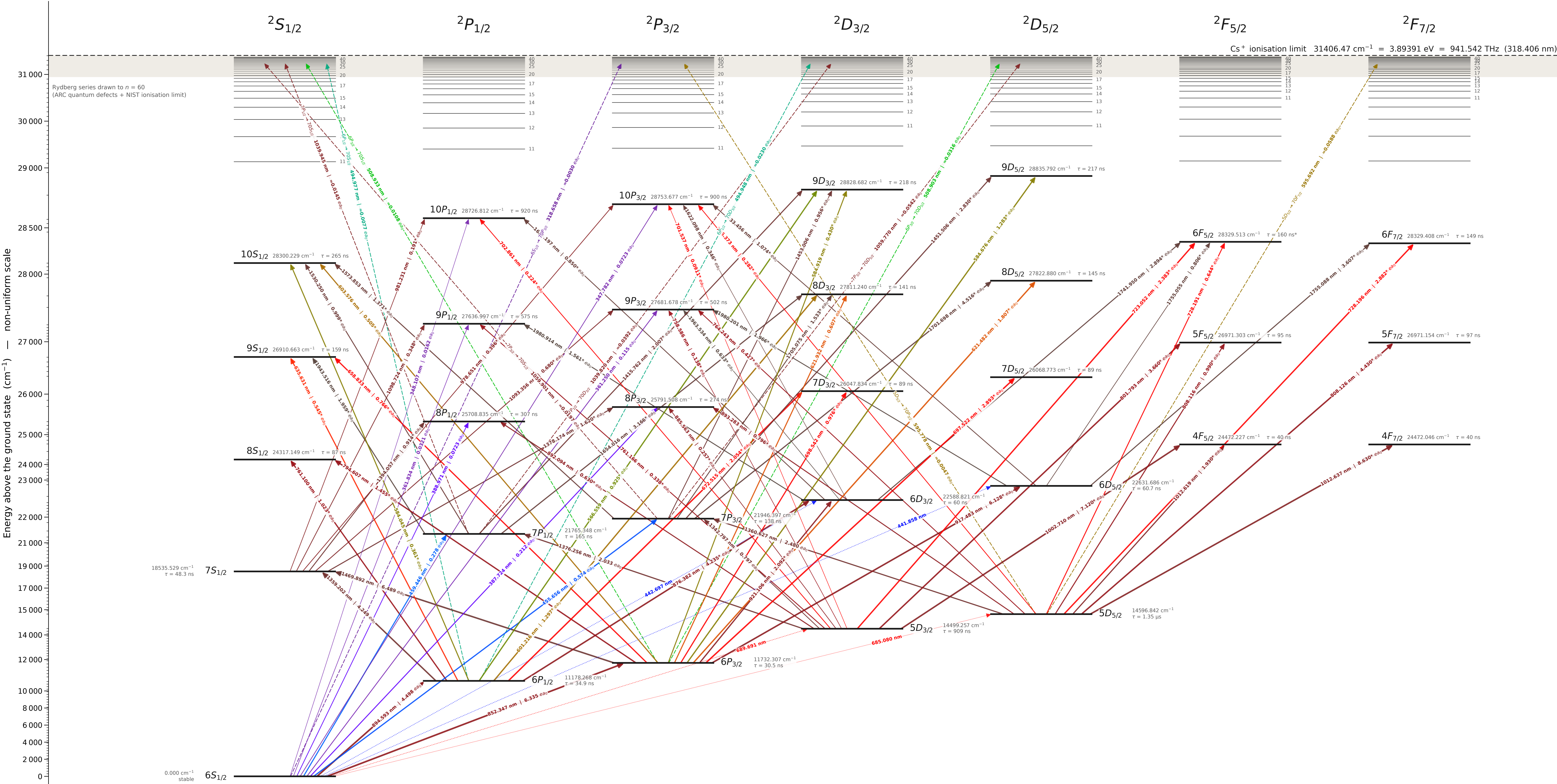
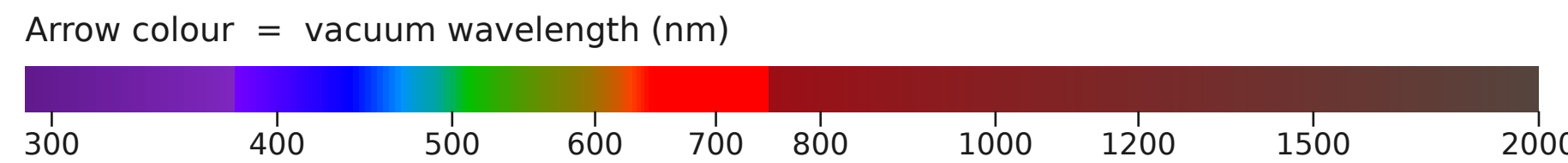


Caesium-133

¹³³Cs (Cs I), nuclear spin $I = 7/2$. Energy levels and transitions below 2 μm: 31 levels, 79 lines.

Arrow label: vacuum wavelength | reduced dipole element $|\langle J||er||J\rangle|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and lifetime (measured where a value exists, otherwise ARC, marked ≈).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\tilde{\nu}$ (cm ⁻¹)	$ \langle J er J\rangle $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Typical use
6S _{1/2} – 6P _{3/2}	852.3473	852.1132	351.7257	11732.307	6.335	3.283e+07	5.225	100.0	D2: MOT, imaging
6S _{1/2} – 6P _{1/2}	894.5930	894.3474	335.1160	11178.268	4.498	2.863e+07	4.556	100.0	D1: optical pumping, Λ / gray-molasses cooling
6S _{1/2} – 7P _{3/2}	455.6557	455.5280	657.9364	21946.397	0.574	1.765e+06	1.157	23.1	blue leg of 456 + 1038 Rydberg excitation
6S _{1/2} – 7P _{1/2}	459.4459	459.3172	652.5087	21765.348	0.278	8.079e+05	0.963	12.5	blue leg (459 + 1040)
6P _{3/2} – 6D _{5/2}	917.4834	917.2317	326.7552	10899.379	6.128*	1.642e+07	2.622	99.6	852 + 917 ladder
6P _{3/2} – 8S _{1/2}	794.6067	794.3882	377.2841	12584.842	1.455*	4.275e+06	1.829	39.6	852 + 795 ladder
6P _{3/2} – 7S _{1/2}	1469.8919	1469.4902	203.9555	6803.221	6.489	1.343e+07	3.296	64.8	telecom E-band ladder; 6S-7S parity-violation line at 539.5 nm x2
6P _{1/2} – 7S _{1/2}	1359.2016	1358.8300	220.5651	7357.260	4.249	7.284e+06	3.296	35.2	telecom O-band ladder
6P _{3/2} – 7D _{5/2}	697.5220	697.3296	429.7964	14336.466	2.893*	8.328e+06	1.788	76.5	852 + 698 ladder
5D _{5/2} – 7P _{3/2}	1360.6267	1360.2547	220.3341	7349.555	2.480	1.237e+06	1.157	20.8	from 5D
5D _{5/2} – 4F _{7/2}	1012.6374	1012.3599	296.0511	9875.203	8.630*	1.817e+07	3.979	93.2	1.01 μm, from 5D
6P _{1/2} – 6D _{3/2}	876.3817	876.1411	342.0798	11410.553	4.235*	1.350e+07	2.653	82.1	895 + 876 ladder
6P _{3/2} – 9S _{1/2}	658.8329	658.6510	455.0356	15178.356	0.766*	2.076e+06	1.001	34.7	852 + 659 ladder
6P _{3/2} – 8D _{5/2}	621.4819	621.3100	482.3832	16090.573	1.807*	4.594e+06	1.098	67.1	852 + 621 ladder

λ and ν from NIST level energies (centre of gravity, no hyperfine structure); λ air uses standard air (Ciddor).
Γ/2π: natural linewidth of the upper level, 1/(2πτ). Branch: share of the upper level's decays that takes this line.

Rydberg excitation wavelengths (vacuum, nm)

Lower → series	n = 30	n = 40	n = 50	n = 70	n = 100	n → ∞	d _∞ ARC
6S _{1/2} → nP _{3/2}	320.005	319.246	318.922	318.658	318.525	318.406	0.0030
6P _{1/2} → nS _{1/2}	498.374	496.443	495.633	494.977	494.651	494.359	0.0077
6P _{1/2} → nD _{3/2}	497.925	496.271	495.550	494.948	494.642	494.359	0.0230
6P _{3/2} → nS _{1/2}	512.526	510.484	509.627	508.934	508.589	508.281	0.0108
6P _{3/2} → nD _{5/2}	512.048	510.301	509.539	508.903	508.579	508.281	0.0316
5D _{3/2} → nP _{3/2}	597.006	594.369	593.249	592.335	591.877	591.464	0.0015
5D _{3/2} → nF _{5/2}	595.770	593.877	593.005	592.249	591.848	591.464	0.0491
5D _{3/2} → nP _{1/2}	600.505	597.836	596.703	595.778	595.315	594.897	0.0047
5D _{5/2} → nF _{7/2}	599.254	597.339	596.457	595.692	595.286	594.897	0.0589
7S _{1/2} → nP _{3/2}	786.536	781.965	780.028	778.448	777.657	776.944	0.0071
7P _{1/2} → nS _{1/2}	1055.057	1046.440	1042.846	1039.945	1038.508	1037.224	0.0145
7P _{1/2} → nD _{3/2}	1053.045	1045.677	1042.478	1039.820	1038.467	1037.224	0.0392
7P _{3/2} → nS _{1/2}	1075.602	1066.648	1062.914	1059.901	1058.408	1057.075	0.0197
7P _{3/2} → nD _{5/2}	1073.501	1065.851	1062.530	1059.770	1058.365	1057.075	0.0542
6D _{5/2} → nP _{3/2}	1160.387	1150.464	1146.276	1142.867	1141.164	1139.630	0.0160
6D _{5/2} → nF _{7/2}	1155.724	1148.622	1145.367	1142.548	1141.057	1139.630	0.0649
8S _{1/2} → nP _{3/2}	1442.513	1427.209	1420.770	1415.536	1412.924	1410.573	0.0131
4F _{7/2} → nD _{5/2}	1472.826	1458.464	1452.253	1447.102	1444.484	1442.081	0.0181

Rydberg term energies from ARC quantum defects + NIST ionisation limit, lower level from NIST. d_∞ = $|\langle J||er||J\rangle|$ to n = 70 in ea₀ (calculated).
S → nP values come from a near-cancelling integral: treat them as an order of magnitude.

¹³³Cs hyperfine structure ($I = 7/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
6S _{1/2}	2298.158	-	F=3: -5170.85 F=4: +4021.78
6P _{1/2}	291.9263(25)	-	F=3: -656.83 F=4: +510.87
6P _{3/2}	50.2878(11)	-0.496(6)	F=2: -339.71 F=3: -188.49 F=4: +12.80 F=5: +263.89
5D _{3/2}	48.78(7)	0.1(7)	F=2: -329.21 F=3: -182.94 F=4: +12.15 F=5: +256.12
5D _{5/2}	-21.24(5)	0.2(5)	F=1: +239.06 F=2: +196.52 F=3: +132.74 F=4: +47.73 F=5: -58.46 F=6: -185.80
7S _{1/2}	545.856(14)	-	F=3: -1228.18 F=4: +955.25
7P _{1/2}	94.37(3)	-	F=3: -212.33 F=4: +165.15
7P _{3/2}	16.605(4)	-0.16(3)	F=2: -112.17 F=3: -62.24 F=4: +4.23 F=5: +87.14
6D _{3/2}	16.338(3)	-0.136(24)	F=2: -110.35 F=3: -61.24 F=4: +4.15 F=5: +85.74
6D _{5/2}	-4.629(11)	-0.10(15)	F=1: +52.02 F=2: +42.79 F=3: +28.94 F=4: +10.45 F=5: -12.70 F=6: -40.53
8S _{1/2}	219.125(3)	-	F=3: -493.03 F=4: +383.47
8P _{1/2}	42.96(9)	-	F=3: -96.66 F=4: +75.18
8P _{3/2}	7.58(3)	-0.14(5)	F=2: -51.24 F=3: -28.40 F=4: +1.96 F=5: +39.76
7D _{3/2}	7.380(8)	-0.17(7)	F=2: 49.91 F=3: -27.64 F=4: +1.92 F=5: +38.70
7D _{5/2}	-1.717(15)	-0.2(5)	F=1: +19.22 F=2: +15.84 F=3: +10.74 F=4: +3.92 F=5: -4.67 F=6: -15.07
8D _{3/2}	3.950(10)	-	F=2: -26.66 F=3: -14.81 F=4: +0.99 F=5: +20.74
8D _{5/2}	-0.85(20)	-	F=1: +9.56 F=2: +7.86 F=3: +5.31 F=4: +1.91 F=5: -2.34 F=6: -7.44

A, B: magnetic-dipole and electric-quadrupole constants (measured; see data file for sources).